United International **University** (**UIU**)

Dept. of Computer Science and Engineering (CSE)

Mid Term Exam Year: **2025** Semester: **Fall**

Course: **CSE 3711/EEE 4413** Title: **Computer Networks (Section – ALL)**

Marks: **30** Time: **1 Hour 30 Minutes**

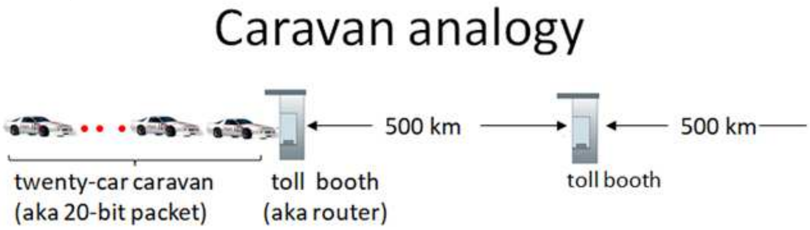
[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

- There are **3 (Three)** questions. Answer **all 3 (Three) questions**. All questions are of values indicated on the right-hand margin.

Q1.

a) Why **Packet Switching** is preferred over **Circuit switching** for today's Internet? **Explain** in terms of growth and efficiency. **[2]**

b) Consider the following **Caravan analogy**:

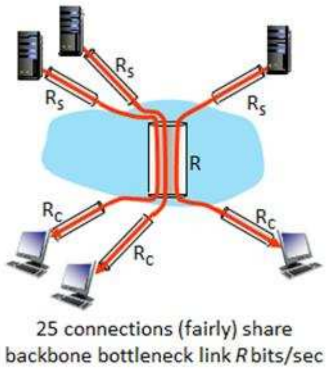


Suppose that the cars propagate at **2000 km/hr** and toll booth takes **2 min** to service a car.

I. Calculate the **transmission** and **propagation delay**. **[2]**

II. **How many cars** will remain at **first booth** if the **first car** arrives to **2nd booth**? **[1]**

c) Assume the **transmission rate** at **client** link **Rc = 50 Mbps**, at **server** link **Rs = 200 Mbps** and **backbone** link **R = 100 Mbps**. Find the **throughput** for the following scenario. Also, find the **transmission time** required to send **1 Gbits data** from a client to server. **[3]**



d) An application on **Host A** sends a **4000-byte message** to **Host B** using the **OSI reference model**. Assume the following:

- Application layer generates **4000 bytes** of data.
- Transport layer uses **TCP** with a **20-byte header**.
- Network layer uses **IP** with a **20-byte header**.
- Data Link layer adds a **14-byte header** and a **4-byte trailer**.
- The data link layer supports a **frame size** of **1518 bytes (including frame header and trailer)**.

I. **Explain how** the **4000-byte application message** is **encapsulated** as it moves down the OSI layers from the Application layer to the Physical layer. **[3]**

II. Determine how many **TCP segments** are required to transmit the message with the **size of the data portion** in each segment. **[1]**

III. Now, if the **frame size** supported by the data link layer is **reduced** to **1200 bytes**, what will be the **change** in the **total overhead**? **Show calculations**. **[2]**

Q2.

a) Consider an **HTTP client** (browser) wants to retrieve a Web document at a given URL. The **IP address** of the **HTTP server** is initially **unknown**. **List**, in **order**, required **transport** and **application layer protocols** for this scenario. **[2]**

b) Suppose within your Web browser you click on a link to obtain a **Web page**. Further, suppose that the Web page associated with the link contains **an HTML file** that references **8 very small objects** on the same server. Let the **RTT** is **0.8 seconds**. Size of each object is **100 Kbytes**. **Throughput** of the network is **500 Kbps**. What is the **time** required to complete to load the **entire web page** using (i) **Non-persistent** and (ii) **Persistent** HTTP connection. **Draw** the corresponding **sequence diagram (steps)** for both the cases. **[3]**

c) I. Classify the following cases with the basic security primitives (Confidentiality, Integrity & Availability) of the CIA triad: **[2]**

- ✓ An attacker succeeded to perform a denial-of-service attack on a server.
- ✓ The receiver detects an altered message sent by the sender during transmission.
- ✓ End-to-end message encryption for an instant messaging application.

II. Give a **1-line explanation** of the following activities **a bad guy (attacker)** can do: ***eavesdropping***, ***impersonation or, spoofing*** and ***hijacking***. **[2]**

III. Illustrate with an example diagram how ***Digital Signature*** supports message ***Integrity*** and **authentication**. **[2]**

Q3. Answer **either (a) or (b) [any one]** from the following:

a) In the following **DNS database** for a university domain (**university.net**), A_{root}, A_{net}, and $A_{university.net}$ denote the **name servers** in the name zone x. **Ax** is a variable and **NOT** a hostname. Suppose you have a **Host C**, a **local name server L**, which is configured with a **root DNS server** with IP address **123.4.5.6**. Assume that all name servers initially have nothing in their caches.

I. Using the resource records given, provide the **hostnames** and **IP addresses** for A_{net}, $A_{university.net}$ **and university's ftp server.** **[2]**

II. Suppose, **Host C** just received the IP address for **www.university.net**. **List the mappings** in the cache of the **local DNS L** and **host C** if iterative or recursive query is used. **[3]**

Name Server Variable	Resource Record
A_{root}	{net, admin.university.net, NS, IN}
A_{root}	{ admin.university.net, 111.5.6.30, A, IN}
A_{com}	{university.net, ns1.university.net, NS, IN}
A_{com}	{ ns1.university.net, 203.0.113.2, A, IN}
$A_{university.net}$	{www.university.net, 203.0.113.10, A, IN}
$A_{university.net}$	{ftp.university.net, 203.0.113.15, A, IN}
$A_{university.net}$	{mail.university.net, 203.0.113.20, A, IN}

OR

b) I. Assume that a **Safari browser** running on a user's host **cse.uiu.ac.bd** requests the URL **www.cisco.com/index.html**. Describe, **step by step**, how the user obtains the **IP address** of www.cisco.com from the local DNS server **dns.uiu.ac.bd** using an **iterated DNS query** in order to send an **HTTP request** to the web server. **Illustrate the query** using a **diagram**. **[3]**

II. **Describe** the **function** of the following **DNS records**: **A, MX, CNAME,** and **NS**. **[2]**

End of Paper – Thank You